Chapter 4: IMU Pre-integration

Derivation of the Three Pre-integrated Increments

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1 Notation and assumptions

We derive the three pre-integrated increments corresponding to the textbook's equations (4.7), starting from the discrete-time IMU kinematics (4.4) and their accumulated form (4.6). Unlike the textbook's constant sampling interval Δt, we allow unequal sample intervals:

$$\boxed{\Delta t_k = t_{k+1} - t_k, \qquad \Delta t_{ij} = t_j - t_i = \sum_{k=i}^{j-1} \Delta t_k.} \tag{1}$$

We assume $R_k \in \mathrm{SO}(3)$ maps body-frame vectors to world-frame vectors, $p_k, v_k, g \in \mathbb{R}^3$ are world-frame quantities, gravity g is constant over the interval, and the bias/noise terms are represented by $b_{g,k}, b_{a,k}, \eta_{gd,k}, \eta_{ad,k}$. The IMU measurements are $\tilde{\omega}_k$ and $\tilde{a}_k$. We use a zero-order-hold approximation over each sample interval for the corrected acceleration and angular velocity. All products of rotations are *time ordered*.

1.1 Continuous-time kinematics (textbook Section 4.1.1)

The starting point is the continuous-time rigid-body kinematics, where $a(t)$ is the *world-frame total acceleration* and $\omega(t)$ is the body-frame angular velocity:

$$\dot{R}(t) = R(t)\omega(t)^{\wedge}, \tag{2a}$$
$$\dot{v}(t) = a(t), \tag{2b}$$
$$\dot{p}(t) = v(t). \tag{2c}$$

On the interval $[t_k, t_{k+1}]$, let $\Delta t_k = t_{k+1} - t_k$. If ω_k is held constant, solving the rotation differential equation gives

$$R_{k+1} = R_k \operatorname{Exp}(\omega_k \Delta t_k). \tag{3}$$

With constant world-frame acceleration a_k over the step, integration of $\dot{v} = a_k$ gives

$$v_{k+1} = v_k + a_k \Delta t_k. \tag{4}$$

Since $v(t_k + \tau) = v_k + a_k \tau$, integrating $\dot{p} = v$ gives

$$\begin{aligned} p_{k+1} &= p_k + \int_0^{\Delta t_k} (v_k + a_k \tau)\, d\tau \\ &= p_k + v_k \Delta t_k + \tfrac{1}{2} a_k \Delta t_k^2. \end{aligned} \tag{5}$$

These are the textbook's single-step discrete kinematics. For the position and velocity steps, holding a_k constant is an approximation: during a real IMU interval the body may rotate, so $R(t)$ and hence the world-frame acceleration can vary.

1.2 Continuous-time IMU measurement model and gravity

Before discretizing the sensor noise, use the continuous-time IMU measurement model of textbook Section 4.1.1. The gyroscope measures body-frame angular velocity, and the accelerometer measures body-frame *specific force* (the acceleration excluding gravity):

$$\tilde{\omega}(t) = \omega(t) + b_g(t) + \eta_g(t), \tag{6a}$$
$$\tilde{a}(t) = R(t)^T \big(a(t) - g\big) + b_a(t) + \eta_a(t). \tag{6b}$$

Here $\eta_g(t)$ and $\eta_a(t)$ denote continuous-time white measurement noise, and $b_g(t), b_a(t)$ are time-dependent biases. Solving for the quantities appearing in the kinematics gives

$$\omega(t) = \tilde{\omega}(t) - b_g(t) - \eta_g(t), \tag{7}$$
$$a(t) = g + R(t)\big(\tilde{a}(t) - b_a(t) - \eta_a(t)\big). \tag{8}$$

Thus gravity appears in the world-frame acceleration even though the accelerometer measures specific force in the body frame.

1.3 Discrete-time noise convention

To apply the continuous-time measurement model over $[t_k, t_{k+1}]$, use the sample interval Δt_k and the interval-averaged discrete noises $\eta_{gd,k}$ and $\eta_{ad,k}$. If $Q_{g,c}$ and $Q_{a,c}$ are the continuous-time white-noise intensity matrices, their discrete sample covariances are

$$\mathrm{Cov}(\eta_{gd,k}) = \frac{Q_{g,c}}{\Delta t_k}, \tag{9a}$$

$$\mathrm{Cov}(\eta_{ad,k}) = \frac{Q_{a,c}}{\Delta t_k}. \tag{9b}$$

These are the discrete measurement-noise samples appearing in the single-step equations below. The factors $1/\Delta t_k$ refer to sample-average noise covariance; the covariance of the integrated noise increment instead scales with Δt_k.

1.4 Single-step discrete IMU propagation

Evaluating the continuous-time measurement model at each sample and using the discrete noise convention above, substitute the corrected angular velocity and world-frame acceleration into the single-step kinematics. This gives the three equations used throughout the derivation:

$$R_{k+1} = R_k \,\mathrm{Exp}((\tilde{\omega}_k - b_{g,k} - \eta_{gd,k})\Delta t_k)\,, \tag{10a}$$

$$v_{k+1} = v_k + g\Delta t_k + R_k(\tilde{a}_k - b_{a,k} - \eta_{ad,k})\Delta t_k, \tag{10b}$$

$$\begin{aligned} p_{k+1} = {} & p_k + v_k\Delta t_k + \tfrac{1}{2}g\Delta t_k^2 \\ & + \tfrac{1}{2}R_k(\tilde{a}_k - b_{a,k} - \eta_{ad,k})\Delta t_k^2. \end{aligned} \tag{10c}$$

The gravity terms follow directly from the world-frame acceleration $a_k = g + R_k(\tilde{a}_k - b_{a,k} - \eta_{ad,k})$. In particular, the factor $\frac{1}{2}$ in the position equation comes from integrating velocity over one sample interval.

2 Rotation increment ΔR_{ij}

Repeated substitution into (10)a gives

$$\begin{aligned} R_{i+1} &= R_i \,\mathrm{Exp}((\tilde{\omega}_i - b_{g,i} - \eta_{gd,i})\Delta t_i), \\ R_j &= R_i \prod_{k=i}^{j-1} \mathrm{Exp}((\tilde{\omega}_k - b_{g,k} - \eta_{gd,k})\Delta t_k). \end{aligned}$$

Left-multiplying by R_i^T and using $R_i^T R_i = I_3$ yields

$$\boxed{\Delta R_{ij} := R_i^T R_j = \prod_{k=i}^{j-1} \mathrm{Exp}((\tilde{\omega}_k - b_{g,k} - \eta_{gd,k})\Delta t_k)\,.} \tag{11}$$

The product is ordered from the earliest sample i on the left to the latest sample $j-1$ on the right; its factors cannot generally be reordered. In particular, $\Delta R_{ii} = I_3$ and

$$\Delta R_{ik} = R_i^T R_k, \qquad R_k = R_i \Delta R_{ik}. \tag{12}$$

The right-hand side of (11) depends on IMU readings, biases, and noise, but not on the starting attitude R_i.

3 Velocity increment Δv_{ij}

Summing (10)b from $k = i$ to $j - 1$ gives a telescoping sum on the left:

$$\begin{aligned} v_j - v_i &= \sum_{k=i}^{j-1} (v_{k+1} - v_k) \\ &= \sum_{k=i}^{j-1} g\Delta t_k + \sum_{k=i}^{j-1} R_k(\tilde{a}_k - b_{a,k} - \eta_{ad,k})\Delta t_k \\ &= g\Delta t_{ij} + \sum_{k=i}^{j-1} R_k(\tilde{a}_k - b_{a,k} - \eta_{ad,k})\Delta t_k. \end{aligned} \tag{13}$$

Substitute $R_k = R_i \Delta R_{ik}$ from (12):

$$\begin{aligned} v_j - v_i - g\Delta t_{ij} &= \sum_{k=i}^{j-1} R_i \Delta R_{ik}(\tilde{a}_k - b_{a,k} - \eta_{ad,k})\Delta t_k \\ &= R_i \sum_{k=i}^{j-1} \Delta R_{ik}(\tilde{a}_k - b_{a,k} - \eta_{ad,k})\Delta t_k. \end{aligned} \tag{14}$$

Left-multiply by R_i^T:

$$\boxed{\begin{aligned} \Delta v_{ij} &:= R_i^T(v_j - v_i - g\Delta t_{ij}) \\ &= \sum_{k=i}^{j-1} \Delta R_{ik}(\tilde{a}_k - b_{a,k} - \eta_{ad,k})\Delta t_k. \end{aligned}} \tag{15}$$

In particular, for any intermediate k,

$$\Delta v_{ik} = R_i^T(v_k - v_i - g\Delta t_{ik}) \quad \Longrightarrow \quad v_k = v_i + g\Delta t_{ik} + R_i \Delta v_{ik}. \tag{16}$$

Thus Δv_{ij} is *not* simply $v_j - v_i$: it removes gravity's contribution and expresses the result in the starting body frame.

4 Position increment Δp_{ij}

4.1 Accumulate the position equation

Summing (10)c gives

$$p_j - p_i = \sum_{k=i}^{j-1} v_k \Delta t_k + \tfrac{1}{2} g \sum_{k=i}^{j-1} \Delta t_k^2 + \tfrac{1}{2} \sum_{k=i}^{j-1} R_k(\tilde{a}_k - b_{a,k} - \eta_{ad,k})\Delta t_k^2. \tag{17}$$

Unlike the velocity case, the first sum contains intermediate velocities v_k. We eliminate them using (16).

4.2 Substitute the intermediate velocity

$$\begin{aligned} \sum_{k=i}^{j-1} v_k \Delta t_k &= \sum_{k=i}^{j-1} (v_i + g\Delta t_{ik} + R_i \Delta v_{ik})\Delta t_k \\ &= v_i \Delta t_{ij} + g \sum_{k=i}^{j-1} \Delta t_{ik} \Delta t_k + R_i \sum_{k=i}^{j-1} \Delta v_{ik} \Delta t_k. \end{aligned} \tag{18}$$

Insert this into (17):

$$p_j - p_i = v_i \Delta t_{ij} + g \left[\sum_{k=i}^{j-1} \Delta t_{ik} \Delta t_k + \tfrac{1}{2} \sum_{k=i}^{j-1} \Delta t_k^2 \right] + R_i \sum_{k=i}^{j-1} \Delta v_{ik} \Delta t_k + \tfrac{1}{2} \sum_{k=i}^{j-1} R_k (\tilde{a}_k - b_{a,k} - \eta_{ad,k}) \Delta t_k^2. \tag{19}$$

4.3 Combine gravity terms using a telescoping sum

Since $\Delta t_{i,k+1} = \Delta t_{ik} + \Delta t_k$,

$$\begin{aligned} \Delta t_{i,k+1}^2 - \Delta t_{ik}^2 &= (\Delta t_{ik} + \Delta t_k)^2 - \Delta t_{ik}^2 \\ &= 2 \Delta t_{ik} \Delta t_k + \Delta t_k^2. \end{aligned} \tag{20}$$

Therefore

$$\Delta t_{ik} \Delta t_k + \tfrac{1}{2} \Delta t_k^2 = \tfrac{1}{2} (\Delta t_{i,k+1}^2 - \Delta t_{ik}^2). \tag{21}$$

Sum over all intervals:

$$\begin{aligned} \sum_{k=i}^{j-1} \left(\Delta t_{ik} \Delta t_k + \tfrac{1}{2} \Delta t_k^2 \right) &= \tfrac{1}{2} \sum_{k=i}^{j-1} (\Delta t_{i,k+1}^2 - \Delta t_{ik}^2) \\ &= \tfrac{1}{2} (\Delta t_{ij}^2 - \Delta t_{ii}^2) \\ &= \boxed{\tfrac{1}{2} \Delta t_{ij}^2}, \qquad \Delta t_{ii} = 0. \end{aligned} \tag{22}$$

This identity holds even when the Δt_k are unequal.

4.4 Factor out the initial rotation

Using $R_k = R_i \Delta R_{ik}$ in (19), and combining the gravity terms, yields

$$p_j - p_i = v_i \Delta t_{ij} + \tfrac{1}{2} g \Delta t_{ij}^2 + R_i \sum_{k=i}^{j-1} \left[\Delta v_{ik} \Delta t_k + \tfrac{1}{2} \Delta R_{ik} (\tilde{a}_k - b_{a,k} - \eta_{ad,k}) \Delta t_k^2 \right]. \tag{23}$$

Move the initial-state terms to the left and multiply by R_i^T:

$$\boxed{\begin{aligned} \Delta p_{ij} &:= R_i^T \left(p_j - p_i - v_i \Delta t_{ij} - \tfrac{1}{2} g \Delta t_{ij}^2 \right) \\ &= \sum_{k=i}^{j-1} \left[\Delta v_{ik} \Delta t_k + \tfrac{1}{2} \Delta R_{ik} (\tilde{a}_k - b_{a,k} - \eta_{ad,k}) \Delta t_k^2 \right]. \end{aligned}} \tag{24}$$

Here Δv_{ik} is the *pre-integrated* velocity increment from i to k, not the world-frame velocity v_k. Its use removes dependence on the initial velocity v_i and initial rotation R_i from the sum.

5 Summary of the three increments

Recall $\Delta t_{ij} = \sum_{k=i}^{j-1} \Delta t_k$. The pre-integrated quantities are

$$\boxed{\Delta R_{ij}} = R_i^T R_j = \prod_{k=i}^{j-1} \mathrm{Exp}((\tilde{\omega}_k - b_{g,k} - \eta_{gd,k})\Delta t_k)\,, \tag{25a}$$

$$\boxed{\Delta v_{ij}} = R_i^T(v_j - v_i - g\Delta t_{ij}) = \sum_{k=i}^{j-1} \Delta R_{ik}(\tilde{a}_k - b_{a,k} - \eta_{ad,k})\Delta t_k, \tag{25b}$$

$$\boxed{\Delta p_{ij}} = R_i^T\left(p_j - p_i - v_i\Delta t_{ij} - \tfrac{1}{2}g\Delta t_{ij}^2\right) = \sum_{k=i}^{j-1}\left(\Delta v_{ik}\Delta t_k + \tfrac{1}{2}\Delta R_{ik}(\tilde{a}_k - b_{a,k} - \eta_{ad,k})\Delta t_k^2\right). \tag{25c}$$

All three right-hand expressions depend on IMU measurements, biases, noise, and sample intervals, not on the absolute initial state (R_i, v_i, p_i). The bias dependence remains; later pre-integration bias corrections address changes in the bias estimates.

6 Interpretation and implementation notes

1. ΔR_{ij} is a relative rotation. Δv_{ij} and Δp_{ij} are gravity-compensated increments expressed in the frame at i; they are not simply $v_j - v_i$ and $p_j - p_i$.
2. Initialization at $k = i$ is $\Delta R_{ii} = I_3$, $\Delta v_{ii} = 0$, and $\Delta p_{ii} = 0$.
3. For unequal sample intervals, replace every single-step Δt by Δt_k, while the accumulated duration is Δt_{ij}.
4. These are the *noise-containing* relative increments of the textbook's definition. Later, the chapter separates noisy quantities from pre-integrated measurements denoted with a tilde and derives their covariance.

7 Pre-integration measurement model through the textbook's (4.18)

7.1 Fixed-bias assumption and noise convention

For the pre-integration interval $[t_i, t_j]$, we assume that the gyroscope and accelerometer biases remain constant and are approximated by their values at the initial time t_i:

$$b_{g,k} \approx b_{g,i}, \qquad b_{a,k} \approx b_{a,i}. \tag{26}$$

Changes in the bias estimates are handled using the bias Jacobians derived in the subsequent Bias Update section. We retain only the following frequently used right-Jacobian notation:

$$J_{r,k} := J_r((\tilde{\omega}_k - b_{g,i})\Delta t_k)\,. \tag{27}$$

Here J_r denotes the right Jacobian of SO(3). The discrete noise samples are $\eta_{gd,k}$ and $\eta_{ad,k}$. The equations below retain only first-order noise terms and neglect products of two noise perturbations.

7.2 Separating rotation measurement and noise

The right-Jacobian first-order identity is

$$\operatorname{Exp}(\theta+\epsilon) \simeq \operatorname{Exp}(\theta)\operatorname{Exp}(J_r(\theta)\epsilon). \tag{28}$$

Setting $\epsilon = -\eta_{gd,k}\Delta t_k$ gives

$$\operatorname{Exp}((\tilde{\omega}_k - b_{g,i} - \eta_{gd,k})\Delta t_k) \simeq \operatorname{Exp}((\tilde{\omega}_k - b_{g,i})\Delta t_k)\operatorname{Exp}(-J_{r,k}\eta_{gd,k}\Delta t_k). \tag{29}$$

Define the noise-free computed measurement

$$\boxed{\Delta\tilde{R}_{ij} := \prod_{k=i}^{j-1} \operatorname{Exp}((\tilde{\omega}_k - b_{g,i})\Delta t_k), \qquad \Delta\tilde{R}_{ii} = I_3.} \tag{30}$$

For any $B \in \mathrm{SO}(3)$, the adjoint identity is

$$\operatorname{Exp}(\xi)B = B\operatorname{Exp}(B^T\xi). \tag{31}$$

For example, the first two increments become

$$\begin{aligned}\Delta R_{i,i+2} &\simeq \Delta\tilde{R}_{i,i+1}\operatorname{Exp}(-J_{r,i}\eta_{gd,i}\Delta t_i)\Delta\tilde{R}_{i+1,i+2}\operatorname{Exp}(-J_{r,i+1}\eta_{gd,i+1}\Delta t_{i+1}) \\ &= \Delta\tilde{R}_{i,i+2}\operatorname{Exp}(-\Delta\tilde{R}_{i+1,i+2}^T J_{r,i}\eta_{gd,i}\Delta t_i)\operatorname{Exp}(-J_{r,i+1}\eta_{gd,i+1}\Delta t_{i+1}).\end{aligned} \tag{32}$$

In particular, the noise accompanying the interval $i+1$ is $\eta_{gd,i+1}$, not $\eta_{gd,i}$; this corrects the index typo in the textbook's displayed (4.10). Repeatedly moving all nominal rotations left gives

$$\Delta R_{ij} \simeq \Delta\tilde{R}_{ij}\prod_{k=i}^{j-1}\operatorname{Exp}\left(-\Delta\tilde{R}_{k+1,j}^T J_{r,k}\eta_{gd,k}\Delta t_k\right). \tag{33}$$

Combining the small noise exponentials to first order, define

$$\boxed{\delta\phi_{ij} := \sum_{k=i}^{j-1}\Delta\tilde{R}_{k+1,j}^T J_{r,k}\eta_{gd,k}\Delta t_k,} \tag{34}$$

where $\Delta\tilde{R}_{jj} = I_3$. Consequently,

$$\boxed{\Delta R_{ij} \simeq \Delta\tilde{R}_{ij}\operatorname{Exp}(-\delta\phi_{ij}).} \tag{35}$$

The adjoint rearrangement (31) is exact; the approximation enters when separating and combining the noise exponentials.

7.3 Velocity measurement and accumulated noise

The relative velocity increment, with fixed accelerometer bias, is

$$\Delta v_{ij} = \sum_{k=i}^{j-1}\Delta R_{ik}((\tilde{a}_k - b_{a,i}) - \eta_{ad,k})\Delta t_k. \tag{36}$$

Using $\Delta R_{ik} \simeq \Delta\tilde{R}_{ik}\operatorname{Exp}(-\delta\phi_{ik})$ and $\operatorname{Exp}(-\delta\phi_{ik}) \simeq I - \delta\phi_{ik}^\wedge$,

$$\begin{aligned}\Delta v_{ij} &\simeq \sum_{k=i}^{j-1}\Delta\tilde{R}_{ik}(I - \delta\phi_{ik}^\wedge)((\tilde{a}_k - b_{a,i}) - \eta_{ad,k})\Delta t_k \\ &\simeq \sum_{k=i}^{j-1}\left[\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i}) + \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge\delta\phi_{ik} - \Delta\tilde{R}_{ik}\eta_{ad,k}\right]\Delta t_k.\end{aligned} \tag{37}$$

The sign of the rotation-noise term is *positive* here, since $-\delta\phi^\wedge f = f^\wedge \delta\phi$. Define

$$\boxed{\Delta\tilde{v}_{ij} := \sum_{k=i}^{j-1} \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})\Delta t_k,} \tag{38}$$

and

$$\boxed{\delta v_{ij} := \sum_{k=i}^{j-1} \left[-\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \delta\phi_{ik} + \Delta\tilde{R}_{ik}\eta_{ad,k}\right] \Delta t_k.} \tag{39}$$

Then

$$\boxed{\Delta v_{ij} \simeq \Delta\tilde{v}_{ij} - \delta v_{ij}.} \tag{40}$$

This sign convention agrees with the textbook's (4.14): the cross-product order determines the apparent sign.

7.4 Position measurement and accumulated noise

Start with the previously derived position increment:

$$\Delta p_{ij} = \sum_{k=i}^{j-1} \left[\Delta v_{ik}\Delta t_k + \tfrac{1}{2}\Delta R_{ik}((\tilde{a}_k - b_{a,i}) - \eta_{ad,k})\Delta t_k^2\right]. \tag{41}$$

Substitute $\Delta v_{ik} \simeq \Delta\tilde{v}_{ik} - \delta v_{ik}$ and $\Delta R_{ik} \simeq \Delta\tilde{R}_{ik}(I - \delta\phi_{ik}^\wedge)$:

$$\begin{aligned}\Delta p_{ij} \simeq \sum_{k=i}^{j-1} \Big[&\Delta\tilde{v}_{ik}\Delta t_k + \tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})\Delta t_k^2 - \delta v_{ik}\Delta t_k \\ &+ \tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \delta\phi_{ik}\Delta t_k^2 - \tfrac{1}{2}\Delta\tilde{R}_{ik}\eta_{ad,k}\Delta t_k^2\Big].\end{aligned} \tag{42}$$

Define the computed pre-integrated position measurement

$$\boxed{\Delta\tilde{p}_{ij} := \sum_{k=i}^{j-1} \left[\Delta\tilde{v}_{ik}\Delta t_k + \tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})\Delta t_k^2\right],} \tag{43}$$

and its accumulated noise

$$\boxed{\begin{aligned}\delta p_{ij} := \sum_{k=i}^{j-1} \Big[&\delta v_{ik}\Delta t_k - \tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \delta\phi_{ik}\Delta t_k^2 \\ &+ \tfrac{1}{2}\Delta\tilde{R}_{ik}\eta_{ad,k}\Delta t_k^2\Big].\end{aligned}} \tag{44}$$

Hence

$$\boxed{\Delta p_{ij} \simeq \Delta\tilde{p}_{ij} - \delta p_{ij}.} \tag{45}$$

8 Summary: pre-integration measurement model

Combining the preceding definitions with the relative-state increments yields the following pre-integration measurement model:

$$\boxed{\Delta\tilde{R}_{ij}} \simeq R_i^T R_j \operatorname{Exp}(\delta\phi_{ij}), \tag{46a}$$

$$\boxed{\Delta\tilde{v}_{ij}} \simeq R_i^T(v_j - v_i - g\Delta t_{ij}) + \delta v_{ij}, \tag{46b}$$

$$\boxed{\Delta\tilde{p}_{ij}} \simeq R_i^T\left(p_j - p_i - v_i\Delta t_{ij} - \tfrac{1}{2}g\Delta t_{ij}^2\right) + \delta p_{ij}. \tag{46c}$$

The positive sign in the rotation measurement is consistent with $\Delta R_{ij} \simeq \Delta\tilde{R}_{ij}\,\mathrm{Exp}(-\delta\phi_{ij})$: right-multiply by $\mathrm{Exp}(\delta\phi_{ij})$ to isolate $\Delta\tilde{R}_{ij}$. Unlike δv_{ij} and δp_{ij}, rotation noise enters multiplicatively on SO(3).

The three tilde quantities are computed from IMU measurements and the fixed bias linearization point. The three delta noise quantities represent accumulated measurement noise, whose covariance is propagated in the following section. The bias-update Jacobians are derived subsequently.

9 Recursive propagation of pre-integration noise

9.1 Conventions and initialization

The accumulated noise is ordered as

$$\eta_{ik} := \begin{bmatrix} \delta\phi_{ik} \\ \delta v_{ik} \\ \delta p_{ik} \end{bmatrix} \in \mathbb{R}^9, \qquad \eta_{d,k} := \begin{bmatrix} \eta_{gd,k} \\ \eta_{ad,k} \end{bmatrix} \in \mathbb{R}^6. \tag{47}$$

We use the fixed biases $b_{g,i}, b_{a,i}$ from the preceding section, and the sample interval $\Delta t_k = t_{k+1} - t_k$. The only additional shorthand is

$$J_{r,k} := J_r((\tilde{\omega}_k - b_{g,i})\Delta t_k)\,. \tag{48}$$

The initial accumulated errors and covariance vanish:

$$\eta_{ii} = 0, \qquad \Sigma_{ii} = 0_{9\times 9}. \tag{49}$$

9.2 Rotation noise recursion

The accumulated rotation perturbation from the preceding section, with the same general interval indices i, j as in the textbook, is

$$\delta\phi_{ij} = \sum_{\ell=i}^{j-1} \Delta\tilde{R}_{\ell+1,j}^T J_{r,\ell}\eta_{gd,\ell}\Delta t_\ell. \tag{50}$$

To derive the recursive update, set $j = k+1$ in this general expression. Separate the final term and use $\Delta\tilde{R}_{\ell+1,k+1} = \Delta\tilde{R}_{\ell+1,k}\Delta\tilde{R}_{k,k+1}$ for $\ell < k$:

$$\delta\phi_{i,k+1} = \sum_{\ell=i}^{k-1} \Delta\tilde{R}_{k,k+1}^T\Delta\tilde{R}_{\ell+1,k}^T J_{r,\ell}\eta_{gd,\ell}\Delta t_\ell + J_{r,k}\eta_{gd,k}\Delta t_k \tag{51}$$

$$= \boxed{\Delta\tilde{R}_{k,k+1}^T\delta\phi_{ik} + J_{r,k}\eta_{gd,k}\Delta t_k}. \tag{52}$$

The transpose of $\Delta\tilde{R}_{k,k+1}$ is the adjoint transport of the previously accumulated rotation noise to the newest right-tangent coordinates.

9.3 Velocity noise recursion

For the general pre-integration interval $[t_i, t_j]$, the accumulated velocity noise is

$$\delta v_{ij} = \sum_{\ell=i}^{j-1} \left[-\Delta\tilde{R}_{i\ell}(\tilde{a}_\ell - b_{a,i})^\wedge\delta\phi_{i\ell}\Delta t_\ell + \Delta\tilde{R}_{i\ell}\eta_{ad,\ell}\Delta t_\ell\right]. \tag{53}$$

To obtain a recursive update, set $j = k+1$ and separate the final IMU interval:

$$\begin{aligned}\delta v_{i,k+1} &= \sum_{\ell=i}^{k-1} \left[-\Delta\tilde{R}_{i\ell}(\tilde{a}_\ell - b_{a,i})^\wedge \delta\phi_{i\ell} + \Delta\tilde{R}_{i\ell}\eta_{ad,\ell} \right] \Delta t_\ell \\ &\quad - \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \delta\phi_{ik}\Delta t_k + \Delta\tilde{R}_{ik}\eta_{ad,k}\Delta t_k \qquad (54) \\ &= \delta v_{ik} - \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \delta\phi_{ik}\Delta t_k + \Delta\tilde{R}_{ik}\eta_{ad,k}\Delta t_k. \qquad (55)\end{aligned}$$

The velocity error therefore couples to the accumulated rotation error.

9.4 Position noise recursion

For the general interval $[t_i, t_j]$, the accumulated position noise is

$$\delta p_{ij} = \sum_{\ell=i}^{j-1} \left[\delta v_{i\ell}\Delta t_\ell - \tfrac{1}{2}\Delta\tilde{R}_{i\ell}(\tilde{a}_\ell - b_{a,i})^\wedge \delta\phi_{i\ell}\Delta t_\ell^2 + \tfrac{1}{2}\Delta\tilde{R}_{i\ell}\eta_{ad,\ell}\Delta t_\ell^2 \right]. \qquad (56)$$

Setting $j = k+1$ and splitting off the final interval yields

$$\delta p_{i,k+1} = \delta p_{ik} + \delta v_{ik}\Delta t_k - \tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \delta\phi_{ik}\Delta t_k^2 + \tfrac{1}{2}\Delta\tilde{R}_{ik}\eta_{ad,k}\Delta t_k^2. \qquad (57)$$

Notice that the δv_{ik} term is the *previous* accumulated velocity noise, consistent with the discrete position integration rule.

9.5 Nine-dimensional matrix recursion

Collecting the preceding three relations gives

$$\boxed{\eta_{i,k+1} = A_k\eta_{ik} + B_k\eta_{d,k}}, \qquad (58)$$

where the 9×9 transition matrix is

$$\boxed{A_k = \begin{bmatrix} \Delta\tilde{R}_{k,k+1}^T & 0 & 0 \\ -\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \Delta t_k & I_3 & 0 \\ -\tfrac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \Delta t_k^2 & I_3\Delta t_k & I_3 \end{bmatrix}} \qquad (59)$$

and the 9×6 noise-input matrix is

$$\boxed{B_k = \begin{bmatrix} J_{r,k}\Delta t_k & 0 \\ 0 & \Delta\tilde{R}_{ik}\Delta t_k \\ 0 & \tfrac{1}{2}\Delta\tilde{R}_{ik}\Delta t_k^2 \end{bmatrix}}. \qquad (60)$$

Rows are ordered $(\delta\phi, \delta v, \delta p)$ and columns of B_k are ordered (η_{gd}, η_{ad}). These blocks are the coefficients of the preceding three recursions, not independent assumptions. We have used the interval index k consistently; the book sometimes labels the last interval $j-1$.

10 Covariance propagation of pre-integrated measurements

10.1 Discrete IMU noise covariance

Let $Q_{d,k} := \mathrm{Cov}(\eta_{d,k})$. If gyroscope and accelerometer discrete noise are independent,

$$Q_{d,k} = \begin{bmatrix} Q_{g,d,k} & 0 \\ 0 & Q_{a,d,k} \end{bmatrix}. \qquad (61)$$

With continuous-time white-noise covariance densities $Q_{g,c}$ and $Q_{a,c}$, the interval-average discrete noise convention used in the textbook gives

$$Q_{g,d,k} = \frac{Q_{g,c}}{\Delta t_k}, \qquad Q_{a,d,k} = \frac{Q_{a,c}}{\Delta t_k}. \tag{62}$$

Thus the covariance of the *noise increments* entering the rotation/velocity equations scales with Δt_k after multiplication by B_k. Do not confuse this $Q_{d,k}$ with the accumulated pre-integration covariance Σ_{ik}.

10.2 Derivation from the recursion

Define $\Sigma_{ik} = \mathrm{Cov}(\eta_{ik})$. Assume the new IMU noise $\eta_{d,k}$ is zero-mean and uncorrelated with the previously accumulated error η_{ik}; treat A_k, B_k as fixed at the current linearization point. Then

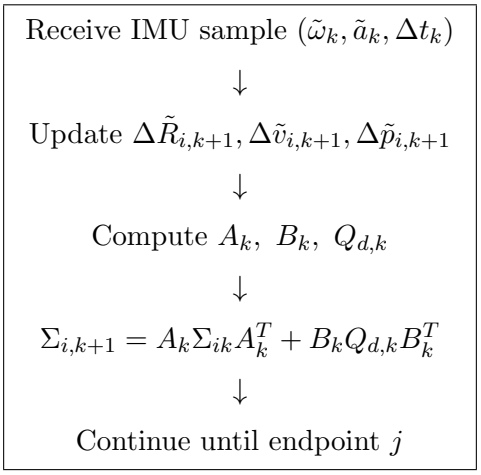
$$\Sigma_{i,k+1} = \mathrm{Cov}(A_k \eta_{ik} + B_k \eta_{d,k}) \tag{63}$$

$$= A_k \Sigma_{ik} A_k^T + B_k Q_{d,k} B_k^T, \tag{64}$$

because the two cross-covariance terms vanish. Therefore,

$$\boxed{\Sigma_{i,k+1} = A_k \Sigma_{ik} A_k^T + B_k Q_{d,k} B_k^T, \qquad \Sigma_{ii} = 0.} \tag{65}$$

10.3 Pre-integration propagation flow



At the endpoint, the computed measurements $(\Delta\tilde{R}_{ij}, \Delta\tilde{v}_{ij}, \Delta\tilde{p}_{ij})$ and their 9×9 covariance Σ_{ij} describe the pre-integrated IMU measurement and its uncertainty. This completes the measurement-noise and covariance propagation of the textbook's Section 4.2.2. The next section begins the bias-update model.

11 Bias Update: First-Order Measurement Correction

11.1 General first-order approximation of a rotation-valued function

Let $R : \mathbb{R}^3 \to \mathrm{SO}(3)$ be a differentiable rotation-valued function of a parameter $\phi \in \mathbb{R}^3$. For a sufficiently small parameter perturbation $\delta\phi$, define its *right-local rotation increment* $f(\delta\phi)$ by

$$\boxed{R(\phi + \delta\phi) = R(\phi)\,\mathrm{Exp}\big(f(\delta\phi)\big).} \tag{66}$$

This is an exact local parameterization of the relative rotation, not yet a first-order approximation. Multiplying on the left by $R(\phi)^T$ and applying the local logarithm gives

$$\boxed{f(\delta\phi) = \mathrm{Log}\big(R(\phi)^T R(\phi + \delta\phi)\big)\,.} \tag{67}$$

Because $f(0) = \mathrm{Log}(I) = 0$, its ordinary Euclidean first-order Taylor expansion at the zero perturbation is

$$\begin{aligned} f(\delta\phi) &= f(0) + \left.\frac{\partial f}{\partial \delta\phi}\right|_{\delta\phi=0} \delta\phi + O(\|\delta\phi\|^2) \\ &= J_{R,\phi}\,\delta\phi + O(\|\delta\phi\|^2), \end{aligned} \tag{68}$$

where the *right-trivialized parameter Jacobian* is defined by

$$\boxed{J_{R,\phi} := \left.\frac{\partial}{\partial \delta\phi} \mathrm{Log}\big(R(\phi)^T R(\phi+\delta\phi)\big)\right|_{\delta\phi=0}.} \tag{69}$$

Substituting into (66) yields the first-order rotation-valued approximation

$$\boxed{R(\phi+\delta\phi) \simeq R(\phi)\,\mathrm{Exp}(J_{R,\phi}\delta\phi).} \tag{70}$$

Here ϕ is a generic *input parameter*; it is not the pre-integration rotation-noise variable $\delta\phi_{ij}$.

11.2 Application to the pre-integrated rotation measurement

In the textbook's bias-update equation, set the generic input parameter to the initial gyroscope-bias estimate $b_{g,i}$. The pre-integrated rotation $\Delta\tilde{R}_{ij}$ is a function of this bias because its IMU increments contain $\tilde{\omega}_k - b_{g,i}$. Thus, applying the right-trivialized Jacobian defined in (69) at the fixed bias $b_{g,i}$ gives

$$\boxed{\Delta\tilde{R}_{ij}(b_{g,i}+\delta b_{g,i}) \simeq \Delta\tilde{R}_{ij}(b_{g,i})\,\mathrm{Exp}\Big(J^{ij}_{\Delta\tilde{R},b_g}\delta b_{g,i}\Big).} \tag{71}$$

Here $J^{ij}_{\Delta\tilde{R},b_g}$ denotes $J_{R,\phi}$ from (69) evaluated for $R(\phi) = \Delta\tilde{R}_{ij}(b_{g,i})$ with $\phi = b_{g,i}$. Equivalently, it is the right-trivialized 3×3 bias Jacobian that the textbook denotes by $\partial\Delta\tilde{R}_{ij}/\partial b_{g,i}$; it is not an entrywise derivative of the rotation matrix. The subsequent bias-Jacobian derivations retain the textbook notation for comparison.

11.3 Velocity and position bias corrections

Unlike the rotation measurement, $\Delta\tilde{v}_{ij}$ and $\Delta\tilde{p}_{ij}$ are vectors in $\mathbb{R}^3$, so ordinary first-order Taylor expansions apply. Let the fixed bias linearization point be $(b_{g,i}, b_{a,i})$ and its correction be $(\delta b_{g,i}, \delta b_{a,i})$. Then

$$\boxed{\begin{aligned} &\Delta\tilde{v}_{ij}(b_{g,i}+\delta b_{g,i}, b_{a,i}+\delta b_{a,i}) \\ &\quad\simeq \Delta\tilde{v}_{ij}(b_{g,i}, b_{a,i}) + \frac{\partial\Delta\tilde{v}_{ij}}{\partial b_{g,i}}\delta b_{g,i} + \frac{\partial\Delta\tilde{v}_{ij}}{\partial b_{a,i}}\delta b_{a,i}. \end{aligned}} \tag{72}$$

Likewise,

$$\boxed{\begin{aligned} &\Delta\tilde{p}_{ij}(b_{g,i}+\delta b_{g,i}, b_{a,i}+\delta b_{a,i}) \\ &\quad\simeq \Delta\tilde{p}_{ij}(b_{g,i}, b_{a,i}) + \frac{\partial\Delta\tilde{p}_{ij}}{\partial b_{g,i}}\delta b_{g,i} + \frac{\partial\Delta\tilde{p}_{ij}}{\partial b_{a,i}}\delta b_{a,i}. \end{aligned}} \tag{73}$$

All five bias Jacobians are evaluated at the fixed bias linearization point. The rotation bias Jacobian $J^{ij}_{\Delta\tilde{R},b_g}$, defined using the right-trivialized derivative in (69), maps a gyroscope bias perturbation to a local rotation vector in $\mathbb{R}^3$. The remaining four Jacobians, associated with the pre-integrated velocity and position, are ordinary Euclidean derivatives.

Equations (71)–(73) summarize the three first-order bias-correction relationships in the textbook's (4.27). The rotation correction uses the exponential map to remain on SO(3), whereas the velocity and position corrections use ordinary vector addition. The five bias Jacobians are derived in the following sections.

11.4 Gyroscope-bias Jacobian: product expansion

We now derive the rotation correction corresponding to the textbook's (4.28)–(4.29), using the same variable sampling intervals Δt_k as in the preceding sections. Throughout the pre-integration interval, $b_{g,i}$ is the fixed bias linearization point, and $\delta b_{g,i}$ is a small change to that *same* bias for every IMU sample.

Start from the pre-integrated rotation as a function of the bias:

$$\Delta\tilde{R}_{ij}(b_{g,i}+\delta b_{g,i}) = \prod_{k=i}^{j-1} \mathrm{Exp}((\tilde{\omega}_k - b_{g,i} - \delta b_{g,i})\Delta t_k)\,. \tag{74}$$

For each sample, the right-Jacobian increment formula from BCH yields

$$\mathrm{Exp}((\tilde{\omega}_k - b_{g,i} - \delta b_{g,i})\Delta t_k) \simeq \mathrm{Exp}((\tilde{\omega}_k - b_{g,i})\Delta t_k)\,\mathrm{Exp}(-J_{r,k}\delta b_{g,i}\Delta t_k). \tag{75}$$

The Jacobian $J_{r,k}$ must be retained because $(\tilde{\omega}_k - b_{g,i})\Delta t_k$ is the nominal increment, not a perturbation that vanishes with $\delta b_{g,i}$.

As in the earlier rotation-noise derivation, use the exact adjoint identity $\mathrm{Exp}(\epsilon)B = B\,\mathrm{Exp}(B^T\epsilon)$ to move each small bias perturbation to the right. First-order BCH then combines these perturbations by addition, neglecting their products, which are $O(\|\delta b_{g,i}\|^2)$. Thus the perturbation introduced at sample k is transported by the subsequent nominal rotation $\Delta\tilde{R}_{k+1,j}^T$.

Repeating the same adjoint transport across the remaining increments, with $\Delta\tilde{R}_{jj} = I_3$, gives

$$\boxed{\begin{aligned}&\Delta\tilde{R}_{ij}(b_{g,i}+\delta b_{g,i})\\ &\quad\simeq \Delta\tilde{R}_{ij}(b_{g,i})\,\mathrm{Exp}\left[-\left(\sum_{k=i}^{j-1}\Delta\tilde{R}_{k+1,j}^T J_{r,k}\Delta t_k\right)\delta b_{g,i}\right].\end{aligned}} \tag{76}$$

This is the first-order rotation bias correction in the textbook's (4.28).

Comparing (76) with the previously defined right-trivialized linearization $\Delta\tilde{R}_{ij}(b+\delta b) \simeq \Delta\tilde{R}_{ij}(b)\,\mathrm{Exp}(\frac{\partial\Delta\tilde{R}_{ij}}{\partial b_{g,i}}\delta b)$ gives

$$\boxed{\frac{\partial\Delta\tilde{R}_{ij}}{\partial b_{g,i}} = -\sum_{k=i}^{j-1}\Delta\tilde{R}_{k+1,j}^T J_{r,k}\Delta t_k.} \tag{77}$$

The derivative notation above refers to the *right-trivialized* 3×3 Jacobian, not an elementwise derivative of a rotation matrix.

11.5 Recursive form of the gyroscope-bias Jacobian

To obtain the textbook's (4.30), separate the last term of (77):

$$\begin{aligned}\frac{\partial\Delta\tilde{R}_{ij}}{\partial b_{g,i}} &= -\sum_{k=i}^{j-2}\Delta\tilde{R}_{k+1,j}^T J_{r,k}\Delta t_k - \Delta\tilde{R}_{j,j}^T J_{r,j-1}\Delta t_{j-1}\\ &= -\sum_{k=i}^{j-2}\Delta\tilde{R}_{k+1,j}^T J_{r,k}\Delta t_k - J_{r,j-1}\Delta t_{j-1}.\end{aligned} \tag{78}$$

For $k \le j-2$, the nominal relative rotations compose as

$$\Delta\tilde{R}_{k+1,j} = \Delta\tilde{R}_{k+1,j-1}\Delta\tilde{R}_{j-1,j}, \qquad \Delta\tilde{R}_{k+1,j}^T = \Delta\tilde{R}_{j-1,j}^T\Delta\tilde{R}_{k+1,j-1}^T. \tag{79}$$

Substituting and factoring out the final rotation gives

$$\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}} = \Delta\tilde{R}_{j-1,j}^T \left[-\sum_{k=i}^{j-2} \Delta\tilde{R}_{k+1,j-1}^T J_{r,k} \Delta t_k \right] - J_{r,j-1}\Delta t_{j-1}$$
$$= \boxed{\Delta\tilde{R}_{j-1,j}^T \frac{\partial \Delta\tilde{R}_{i,j-1}}{\partial b_{g,i}} - J_{r,j-1}\Delta t_{j-1}}. \tag{80}$$

Equivalently, for an incoming IMU interval $[t_k, t_{k+1}]$,

$$\boxed{\frac{\partial \Delta\tilde{R}_{i,k+1}}{\partial b_{g,i}} = \Delta\tilde{R}_{k,k+1}^T \frac{\partial \Delta\tilde{R}_{ik}}{\partial b_{g,i}} - J_{r,k}\Delta t_k, \qquad \frac{\partial \Delta\tilde{R}_{ii}}{\partial b_{g,i}} = 0_{3\times 3}.} \tag{81}$$

The first term transports the previously accumulated Jacobian to the new rotation reference, and the second adds the current interval's bias sensitivity. In the textbook's (4.30), the last $J_{r,k}\Delta t$ should be read as the contribution of the *last* interval, $J_{r,j-1}\Delta t_{j-1}$, when the left-hand side ends at j.

12 Velocity and Position Bias Jacobians

We now complete the five bias Jacobians of the pre-integrated measurements, corresponding to the textbook's (4.33)–(4.34). All Jacobians are evaluated at the fixed bias linearization point $(b_{g,i}, b_{a,i})$. We retain the sampling interval Δt_k.

12.1 Velocity with respect to accelerometer bias

From the measurement definition,

$$\Delta\tilde{v}_{ij} = \sum_{k=i}^{j-1} \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})\Delta t_k. \tag{82}$$

The rotation increment does not depend on accelerometer bias. Thus

$$\boxed{\frac{\partial \Delta\tilde{v}_{ij}}{\partial b_{a,i}} = -\sum_{k=i}^{j-1} \Delta\tilde{R}_{ik}\Delta t_k.} \tag{83}$$

12.2 Velocity with respect to gyroscope bias

The gyro bias affects the velocity through the pre-integrated rotation. Apply the right-trivialized perturbation derived earlier:

$$\Delta\tilde{R}_{ik}(b_{g,i} + \delta b_g) \simeq \Delta\tilde{R}_{ik}\,\mathrm{Exp}\Big(\frac{\partial \Delta\tilde{R}_{ik}}{\partial b_{g,i}}\delta b_g\Big). \tag{84}$$

Keeping only first-order terms,

$$\Delta\tilde{v}_{ij}(b_{g,i} + \delta b_g) \simeq \sum_{k=i}^{j-1} \Delta\tilde{R}_{ik} \left[I + \Big(\frac{\partial \Delta\tilde{R}_{ik}}{\partial b_{g,i}}\delta b_g\Big)^\wedge \right] (\tilde{a}_k - b_{a,i})\Delta t_k$$
$$= \Delta\tilde{v}_{ij}(b_{g,i}) - \sum_{k=i}^{j-1} \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \frac{\partial \Delta\tilde{R}_{ik}}{\partial b_{g,i}}\delta b_g \Delta t_k. \tag{85}$$

Here $u^\wedge w = -w^\wedge u$. Comparing with the first-order Taylor expansion gives

$$\boxed{\frac{\partial \Delta\tilde{v}_{ij}}{\partial b_{g,i}} = -\sum_{k=i}^{j-1} \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \frac{\partial \Delta\tilde{R}_{ik}}{\partial b_{g,i}}\Delta t_k.} \tag{86}$$

12.3 Position with respect to accelerometer bias

Recall

$$\Delta\tilde{p}_{ij} = \sum_{k=i}^{j-1} \left[\Delta\tilde{v}_{ik}\Delta t_k + \frac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})\Delta t_k^2 \right]. \tag{87}$$

Differentiate both terms. The rotation has no dependence on $b_{a,i}$:

$$\boxed{\frac{\partial\Delta\tilde{p}_{ij}}{\partial b_{a,i}} = \sum_{k=i}^{j-1} \left[\frac{\partial\Delta\tilde{v}_{ik}}{\partial b_{a,i}}\Delta t_k - \frac{1}{2}\Delta\tilde{R}_{ik}\Delta t_k^2 \right].} \tag{88}$$

12.4 Position with respect to gyroscope bias

There are two dependency paths: gyro bias affects the accumulated velocity $\Delta\tilde{v}_{ik}$ and the rotation $\Delta\tilde{R}_{ik}$ in the acceleration contribution. Under $b_{g,i} \mapsto b_{g,i} + \delta b_g$, insert

$$\Delta\tilde{v}_{ik}(b_{g,i} + \delta b_g) \simeq \Delta\tilde{v}_{ik} + \frac{\partial\Delta\tilde{v}_{ik}}{\partial b_{g,i}}\delta b_g, \tag{89}$$

$$\Delta\tilde{R}_{ik}(b_{g,i} + \delta b_g) \simeq \Delta\tilde{R}_{ik} \left[I + (\frac{\partial\Delta\tilde{R}_{ik}}{\partial b_{g,i}}\delta b_g)^\wedge \right]. \tag{90}$$

After collecting the terms linear in δb_g,

$$\begin{aligned}\Delta\tilde{p}_{ij}(b_{g,i} + \delta b_g) \simeq\ & \Delta\tilde{p}_{ij}(b_{g,i}) \\ & + \sum_{k=i}^{j-1} \left[\frac{\partial\Delta\tilde{v}_{ik}}{\partial b_{g,i}}\Delta t_k - \frac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \frac{\partial\Delta\tilde{R}_{ik}}{\partial b_{g,i}}\Delta t_k^2 \right] \delta b_g.\end{aligned} \tag{91}$$

Therefore,

$$\boxed{\frac{\partial\Delta\tilde{p}_{ij}}{\partial b_{g,i}} = \sum_{k=i}^{j-1} \left[\frac{\partial\Delta\tilde{v}_{ik}}{\partial b_{g,i}}\Delta t_k - \frac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \frac{\partial\Delta\tilde{R}_{ik}}{\partial b_{g,i}}\Delta t_k^2 \right].} \tag{92}$$

13 Recursive Forms of the Five Bias Jacobians

For a new IMU interval $[t_k, t_{k+1}]$, split the last term from each sum. The rotation Jacobian has already been derived. The complete set of recursions is

$$\frac{\partial\Delta\tilde{R}_{i,k+1}}{\partial b_{g,i}} = \Delta\tilde{R}_{k,k+1}^T \frac{\partial\Delta\tilde{R}_{ik}}{\partial b_{g,i}} - J_{r,k}\Delta t_k, \tag{93}$$

$$\frac{\partial\Delta\tilde{v}_{i,k+1}}{\partial b_{a,i}} = \frac{\partial\Delta\tilde{v}_{ik}}{\partial b_{a,i}} - \Delta\tilde{R}_{ik}\Delta t_k, \tag{94}$$

$$\frac{\partial\Delta\tilde{v}_{i,k+1}}{\partial b_{g,i}} = \frac{\partial\Delta\tilde{v}_{ik}}{\partial b_{g,i}} - \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \frac{\partial\Delta\tilde{R}_{ik}}{\partial b_{g,i}}\Delta t_k, \tag{95}$$

$$\frac{\partial\Delta\tilde{p}_{i,k+1}}{\partial b_{a,i}} = \frac{\partial\Delta\tilde{p}_{ik}}{\partial b_{a,i}} + \frac{\partial\Delta\tilde{v}_{ik}}{\partial b_{a,i}}\Delta t_k - \frac{1}{2}\Delta\tilde{R}_{ik}\Delta t_k^2, \tag{96}$$

$$\begin{aligned}\frac{\partial\Delta\tilde{p}_{i,k+1}}{\partial b_{g,i}} =\ & \frac{\partial\Delta\tilde{p}_{ik}}{\partial b_{g,i}} + \frac{\partial\Delta\tilde{v}_{ik}}{\partial b_{g,i}}\Delta t_k \\ & - \frac{1}{2}\Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})^\wedge \frac{\partial\Delta\tilde{R}_{ik}}{\partial b_{g,i}}\Delta t_k^2.\end{aligned} \tag{97}$$

All five Jacobians are initially zero at $k = i$. The right-hand sides use the *old* values at k, including $\Delta\tilde{R}_{ik}$ and the old velocity and rotation Jacobians. In an in-place implementation, update the position Jacobians first, then the velocity Jacobians, and finally the rotation Jacobian, or cache all old quantities.

13.1 Summary

The rotation bias Jacobian is a right-trivialized SO(3) derivative, whereas the other four bias Jacobians are ordinary Euclidean derivatives. Equations (93)–(97) provide recursive updates for the sensitivities of the pre-integrated rotation, velocity, and position measurements with respect to the gyroscope and accelerometer biases.

These Jacobians enable efficient first-order bias correction without re-integrating the original IMU measurements whenever the bias estimates change during graph optimization. Together with the pre-integrated measurements and their covariance, they provide the information needed to construct the IMU pre-integration factor in the following section.

14 Pre-integration Factor in Graph Optimization

14.1 Purpose and keyframe states

The pre-integration measurement model derived above can now be used as an IMU constraint between keyframes i and j. At each keyframe, we use the state variables

$$x_k = (R_k, p_k, v_k, b_{a,k}, b_{g,k}). \tag{98}$$

The purpose of this section is to define the *IMU pre-integration factor*, not yet to assemble a complete multi-sensor optimization problem. The pre-integrated measurements are $\Delta\tilde{R}_{ij}$, $\Delta\tilde{v}_{ij}$ and $\Delta\tilde{p}_{ij}$, with covariance Σ_{ij}.

14.2 Residual definition

The measurement model gives a predicted relative motion from the keyframe states. The residual compares this prediction against the pre-integrated IMU measurement. Following the textbook's residual convention, define

$$r_{\Delta R_{ij}} = \mathrm{Log}\left(\Delta\tilde{R}_{ij}^T R_i^T R_j\right), \tag{99a}$$

$$r_{\Delta v_{ij}} = R_i^T(v_j - v_i - g\Delta t_{ij}) - \Delta\tilde{v}_{ij}, \tag{99b}$$

$$r_{\Delta p_{ij}} = R_i^T\left(p_j - p_i - v_i\Delta t_{ij} - \tfrac{1}{2}g\Delta t_{ij}^2\right) - \Delta\tilde{p}_{ij}. \tag{99c}$$

The rotation residual uses the $SO(3)$ logarithm instead of subtracting rotation matrices. When the state-predicted relative motion agrees with the pre-integrated measurements, all three residuals vanish.

Stack the three components in a nine-dimensional vector:

$$\boxed{r_{ij}^{\mathrm{IMU}} = \begin{bmatrix} r_{\Delta R_{ij}} \\ r_{\Delta v_{ij}} \\ r_{\Delta p_{ij}} \end{bmatrix} \in \mathbb{R}^9.} \tag{100}$$

The pre-integration covariance derived previously supplies the weighting for a least-squares IMU factor, schematically

$$E_{ij}^{\mathrm{IMU}} = \left(r_{ij}^{\mathrm{IMU}}\right)^T \Sigma_{ij}^{-1} r_{ij}^{\mathrm{IMU}}. \tag{101}$$

This is the standard weighted least-squares interpretation of the textbook's residual and covariance; the section primarily defines the residuals and factor connectivity.

14.3 States, measurements, and residuals

These are three distinct mathematical objects. The keyframe states x_i, x_j are unknown optimization variables (15 degrees of freedom per keyframe). The pre-integrated IMU measurements $(\Delta\tilde{R}_{ij}, \Delta\tilde{v}_{ij}, \Delta\tilde{p}_{ij})$ are obtained from IMU samples at a fixed bias linearization point; they have nine local degrees of freedom, but are *not* navigation states being optimized. The measurement residual $r_{ij}^{\text{IMU}} \in \mathbb{R}^9$ compares the motion predicted from the current keyframe states with these measurements.

A factor-graph connection means that a residual *depends on* a state variable; it does not imply that the IMU directly measures that variable. The dependencies, including first-order bias correction, are

$$\begin{aligned} r_{\Delta R_{ij}} &= r_{\Delta R_{ij}}(R_i, R_j, b_{g,i}), \\ r_{\Delta v_{ij}} &= r_{\Delta v_{ij}}(R_i, v_i, v_j, b_{g,i}, b_{a,i}), \\ r_{\Delta p_{ij}} &= r_{\Delta p_{ij}}(R_i, p_i, p_j, v_i, b_{g,i}, b_{a,i}). \end{aligned}$$

The endpoint biases $b_{a,j}, b_{g,j}$ do not enter this nine-dimensional pre-integration residual. Instead, separate bias random-walk factors constrain their changes, for example

$$r_{b_a} = b_{a,j} - b_{a,i}, \qquad r_{b_g} = b_{g,j} - b_{g,i}. \tag{102}$$

These two factors each have three residual components.

14.4 Jacobian dependency matrix

The rows below represent *measurement residuals*, not pre-integrated measurements; the columns represent local perturbations of optimization states. A check mark indicates a generally nonzero 3×3 Jacobian block. A dash denotes a zero block.

	ϕ_i	δp_i	δv_i	$\delta b_{a,i}$	$\delta b_{g,i}$	ϕ_j	δp_j	δv_j	$\delta b_{a,j}$	$\delta b_{g,j}$
$r_{\Delta R_{ij}}$	✓	—	—	—	✓	✓	—	—	—	—
$r_{\Delta v_{ij}}$	✓	—	✓	✓	✓	—	—	✓	—	—
$r_{\Delta p_{ij}}$	✓	✓	✓	✓	✓	—	✓	—	—	—

Here $R_i' = R_i \operatorname{Exp}(\phi_i)$ and $R_j' = R_j \operatorname{Exp}(\phi_j)$. For the complete two-keyframe local perturbation vector $\delta X_{ij} \in \mathbb{R}^{30}$, the residual Jacobian is

$$J_{\text{pre}} = \frac{\partial r_{ij}^{\text{IMU}}}{\partial \delta X_{ij}} \in \mathbb{R}^{9\times 30}. \tag{103}$$

The final two bias columns are zero. Restricting to the eight three-dimensional state blocks on which the pre-integration residual depends gives a 9×24 Jacobian. Its assembled block form is recorded in the Appendix; the main text develops the individual Jacobian blocks directly from the residual definitions.

14.5 Dependence on the IMU biases

Although the residual expressions explicitly display only the rotation, position and velocity states, their pre-integrated measurement terms depend on the biases at the starting keyframe:

$$\Delta\tilde{R}_{ij} = \Delta\tilde{R}_{ij}(b_{g,i}), \tag{104}$$

$$\Delta\tilde{v}_{ij} = \Delta\tilde{v}_{ij}(b_{g,i}, b_{a,i}), \tag{105}$$

$$\Delta\tilde{p}_{ij} = \Delta\tilde{p}_{ij}(b_{g,i}, b_{a,i}). \tag{106}$$

The five bias Jacobians derived in the preceding section provide first-order corrections to these measurements when the optimization updates the biases, without reintegrating all IMU samples. The IMU pre-integration factor therefore connects $(R_i, p_i, v_i, b_{a,i}, b_{g,i})$ to (R_j, p_j, v_j). The bias random walks between the two keyframes are represented by separate bias factors, as illustrated in the textbook's Figure 4-1.

14.6 Scope: IMU constraints versus other sensor factors

At this stage, we have formulated *only* the IMU pre-integration constraint and the associated bias random-walk model. We have not yet introduced GNSS, wheel-encoder or other sensor observation factors into the graph. The IMU factor constrains relative motion, rather than providing an absolute position or attitude reference by itself.

Later, a full graph may combine IMU pre-integration, bias random walk, GNSS, odometry and prior factors. The precise graph construction and implementation are addressed in later sections of the chapter.

14.7 Summary

The conceptual progression is

$$\boxed{\text{IMU samples} \;\longrightarrow\; (\Delta\tilde{R}_{ij}, \Delta\tilde{v}_{ij}, \Delta\tilde{p}_{ij}, \Sigma_{ij}) \;\longrightarrow\; r_{ij}^{\text{IMU}} \;\longrightarrow\; \text{IMU factor.}}$$

The following sections compute the residual Jacobians with respect to the graph's optimization variables.

15 Jacobians of Preintegration Residuals

15.1 Rotation residual: right perturbations

We now linearize the rotation residual of the IMU factor with respect to its endpoint rotations. Following the same convention as the preceding sections, perturb each rotation on the right:

$$R_i' = R_i \operatorname{Exp}(\phi_i), \qquad R_j' = R_j \operatorname{Exp}(\phi_j), \tag{107}$$

where $\phi_i, \phi_j \in \mathbb{R}^3$ are small optimization increments, not the pre-integration rotation noise $\delta\phi_{ij}$. Recall the residual

$$r_{\Delta R_{ij}} = \operatorname{Log}\left(\Delta\tilde{R}_{ij}^T R_i^T R_j\right). \tag{108}$$

The pre-integrated measurement is held fixed when differentiating with respect to the endpoint rotations.

15.2 Derivative with respect to the initial rotation

Perturb R_i while holding R_j fixed:

$$\begin{aligned} r_{\Delta R_{ij}}(R_i \operatorname{Exp}(\phi_i)) &= \operatorname{Log}\Big(\Delta\tilde{R}_{ij}^T (R_i \operatorname{Exp}(\phi_i))^T R_j\Big) \\ &= \operatorname{Log}\Big(\Delta\tilde{R}_{ij}^T \operatorname{Exp}(-\phi_i) R_i^T R_j\Big). \end{aligned} \tag{109}$$

The SO(3) adjoint identity

$$\operatorname{Exp}(\xi) Q = Q \operatorname{Exp}(Q^T \xi), \qquad Q \in \mathrm{SO}(3), \tag{110}$$

with $Q = R_i^T R_j$, moves the small perturbation to the right:

$$\operatorname{Exp}(-\phi_i) R_i^T R_j = R_i^T R_j \operatorname{Exp}(-R_j^T R_i \phi_i), \tag{111}$$

$$r_{\Delta R_{ij}}(R_i \operatorname{Exp}(\phi_i)) = \operatorname{Log}\Big(\Delta\tilde{R}_{ij}^T R_i^T R_j \operatorname{Exp}(-R_j^T R_i \phi_i)\Big). \tag{112}$$

By (108), $\Delta\tilde{R}_{ij}^T R_i^T R_j = \operatorname{Exp}(r_{\Delta R_{ij}})$. Use the first-order right-BCH identity

$$\operatorname{Log}(\operatorname{Exp}(a)\operatorname{Exp}(\epsilon)) \simeq a + J_r^{-1}(a)\epsilon, \qquad \epsilon \text{ small}. \tag{113}$$

Consequently,

$$\begin{aligned} r_{\Delta R_{ij}}(R_i \operatorname{Exp}(\phi_i)) &\simeq \operatorname{Log}\big(\operatorname{Exp}(r_{\Delta R_{ij}}) \operatorname{Exp}(-R_j^T R_i \phi_i)\big) \\ &\simeq r_{\Delta R_{ij}} - J_r^{-1}(r_{\Delta R_{ij}}) R_j^T R_i \phi_i. \end{aligned} \tag{114}$$

Comparing with a first-order Taylor expansion yields

$$\boxed{\frac{\partial r_{\Delta R_{ij}}}{\partial \phi_i} = -J_r^{-1}(r_{\Delta R_{ij}}) R_j^T R_i.} \tag{115}$$

The minus sign comes from $(R_i \operatorname{Exp}(\phi_i))^T = \operatorname{Exp}(-\phi_i) R_i^T$; the $R_j^T R_i$ term comes from the adjoint transformation.

15.3 Derivative with respect to the final rotation

Perturb R_j while holding R_i fixed:

$$\begin{aligned} r_{\Delta R_{ij}}(R_j \operatorname{Exp}(\phi_j)) &= \operatorname{Log}\Big(\Delta\tilde{R}_{ij}^T R_i^T R_j \operatorname{Exp}(\phi_j)\Big) \\ &= \operatorname{Log}\big(\operatorname{Exp}(r_{\Delta R_{ij}}) \operatorname{Exp}(\phi_j)\big) \\ &\simeq r_{\Delta R_{ij}} + J_r^{-1}(r_{\Delta R_{ij}}) \phi_j. \end{aligned} \tag{116}$$

Here the perturbation is already at the far right, so no adjoint rearrangement is required. Therefore,

$$\boxed{\frac{\partial r_{\Delta R_{ij}}}{\partial \phi_j} = J_r^{-1}(r_{\Delta R_{ij}}).} \tag{117}$$

15.4 Summary of the two rotation-residual Jacobians

$$\frac{\partial r_{\Delta R_{ij}}}{\partial \phi_i} = -J_r^{-1}(r_{\Delta R_{ij}}) R_j^T R_i, \tag{118}$$

$$\frac{\partial r_{\Delta R_{ij}}}{\partial \phi_j} = J_r^{-1}(r_{\Delta R_{ij}}). \tag{119}$$

Both are 3×3 Jacobians with respect to right perturbations. These are the endpoint-rotation derivatives corresponding to the textbook's (4.37)–(4.38). The following sections extend these results to the bias, velocity, and position residual Jacobians.

16 Remaining IMU-Factor Residual Jacobians

16.1 Rotation residual with respect to gyroscope bias

Let $b_{g,i}$ denote the original bias linearization point. The optimizer has already applied the correction $\delta b_{g,i}$, so its current bias estimate is

$$b'_g = b_{g,i} + \delta b_{g,i}. \tag{120}$$

To compute the residual Jacobian, we perturb this *current* estimate by an additional infinitesimal increment ϵ_g:

$$b''_g = b'_g + \epsilon_g = b_{g,i} + \delta b_{g,i} + \epsilon_g. \tag{121}$$

The important distinction is that the original pre-integration and its bias Jacobian $\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}$ were calculated at $b_{g,i}$. We do *not* recompute either of them at b'_g. Instead, we differentiate the first-order bias-correction model constructed at the original linearization point. Retain $\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}$ as defined earlier and set

$$J_{r,b} := J_r\big(\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}\delta b_{g,i}\big). \tag{122}$$

The currently corrected measurement and residual are

$$\Delta\tilde{R}'_{ij} = \Delta\tilde{R}_{ij}\,\mathrm{Exp}\big(\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}\delta b_{g,i}\big), \tag{123}$$

$$r'_{\Delta R_{ij}} = \mathrm{Log}\big((\Delta\tilde{R}'_{ij})^T R_i^T R_j\big). \tag{124}$$

Although ϵ_g perturbs the current estimate b'_g, the fixed-linearization correction model still expands the total bias offset $\delta b_{g,i} + \epsilon_g$ around $b_{g,i}$:

$$\Delta\tilde{R}_{ij}(b'_g + \epsilon_g) \simeq \Delta\tilde{R}_{ij}(b_{g,i})\,\mathrm{Exp}\big(\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}(\delta b_{g,i} + \epsilon_g)\big). \tag{125}$$

Now use the first-order BCH formula with $\phi = \frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}\delta b_{g,i}$ and $\delta\phi = \frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}\epsilon_g$ to separate the new perturbation:

$$\begin{aligned}
&\Delta\tilde{R}_{ij}(b_{g,i} + \delta b_{g,i} + \epsilon_g) \\
&\quad\simeq \Delta\tilde{R}_{ij}\,\mathrm{Exp}\big(\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}(\delta b_{g,i} + \epsilon_g)\big) \\
&\quad\simeq \Delta\tilde{R}'_{ij}\,\mathrm{Exp}(J_{r,b}\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}\epsilon_g).
\end{aligned} \tag{126}$$

Here $J_{r,b} = J_r(\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}\delta b_{g,i})$ is the right Jacobian of the *bias-induced rotation correction*. The resulting local sensitivity $J_{r,b}\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}$ is the derivative of the *fixed first-order correction model* at the current correction $\delta b_{g,i}$. It need not equal the Jacobian obtained by recomputing the original IMU pre-integration at b'_g; the latter would account for higher-order bias dependence.

Substitute this into the rotation residual and use $(AB)^T = B^T A^T$:

$$\begin{aligned} r_{\Delta R_{ij}}(\epsilon_g) &\simeq \mathrm{Log}\left(\mathrm{Exp}(-J_{r,b}\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}\epsilon_g)(\Delta\tilde{R}'_{ij})^T R_i^T R_j\right) \\ &= \mathrm{Log}\left(\mathrm{Exp}(-J_{r,b}\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}\epsilon_g)\,\mathrm{Exp}(r'_{\Delta R_{ij}})\right) \\ &= \mathrm{Log}\left(\mathrm{Exp}(r'_{\Delta R_{ij}})\,\mathrm{Exp}\big(-\mathrm{Exp}(r'_{\Delta R_{ij}})^T J_{r,b}\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}\epsilon_g\big)\right) \\ &\simeq r'_{\Delta R_{ij}} - J_r^{-1}(r'_{\Delta R_{ij}})\,\mathrm{Exp}(r'_{\Delta R_{ij}})^T J_{r,b}\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}\epsilon_g. \end{aligned} \tag{127}$$

The third line uses the exact SO(3) adjoint identity; the last line uses the first-order right-BCH formula. Thus

$$\boxed{\frac{\partial r_{\Delta R_{ij}}}{\partial \epsilon_g} = -J_r^{-1}(r'_{\Delta R_{ij}})\,\mathrm{Exp}(r'_{\Delta R_{ij}})^T J_{r,b}\frac{\partial \Delta\tilde{R}_{ij}}{\partial b_{g,i}}.} \tag{128}$$

16.2 Velocity residual Jacobians

Recall the velocity residual directly, without introducing an additional shorthand for the state-predicted velocity:

$$r_{\Delta v_{ij}} = R_i^T(v_j - v_i - g\Delta t_{ij}) - \Delta\tilde{v}_{ij}. \tag{129}$$

Its Euclidean derivatives are

$$\boxed{\frac{\partial r_{\Delta v_{ij}}}{\partial v_i} = -R_i^T, \qquad \frac{\partial r_{\Delta v_{ij}}}{\partial v_j} = R_i^T.} \tag{130}$$

Using the right rotation perturbation $R'_i = R_i\,\mathrm{Exp}(\phi_i)$,

$$\begin{aligned} r_{\Delta v_{ij}}(R'_i) &= \mathrm{Exp}(-\phi_i)R_i^T(v_j - v_i - g\Delta t_{ij}) - \Delta\tilde{v}_{ij} \\ &\simeq r_{\Delta v_{ij}} + \big[R_i^T(v_j - v_i - g\Delta t_{ij})\big]^\wedge \phi_i. \end{aligned} \tag{131}$$

Consequently,

$$\boxed{\frac{\partial r_{\Delta v_{ij}}}{\partial \phi_i} = \big[R_i^T(v_j - v_i - g\Delta t_{ij})\big]^\wedge.} \tag{132}$$

Since the pre-integrated measurement is subtracted, its bias Jacobians enter with negative signs:

$$\frac{\partial r_{\Delta v_{ij}}}{\partial b_{g,i}} = -\frac{\partial \Delta\tilde{v}_{ij}}{\partial b_{g,i}}, \qquad \frac{\partial r_{\Delta v_{ij}}}{\partial b_{a,i}} = -\frac{\partial \Delta\tilde{v}_{ij}}{\partial b_{a,i}}. \tag{133}$$

16.3 Position residual Jacobians

The position residual is

$$r_{\Delta p_{ij}} = R_i^T\left(p_j - p_i - v_i\Delta t_{ij} - \tfrac{1}{2}g\Delta t_{ij}^2\right) - \Delta\tilde{p}_{ij}. \tag{134}$$

The Euclidean derivatives are

$$\frac{\partial r_{\Delta p_{ij}}}{\partial p_i} = -R_i^T, \qquad \frac{\partial r_{\Delta p_{ij}}}{\partial p_j} = R_i^T, \qquad \frac{\partial r_{\Delta p_{ij}}}{\partial v_i} = -R_i^T\Delta t_{ij}. \tag{135}$$

With $R_i' = R_i \operatorname{Exp}(\phi_i)$,

$$\begin{aligned} r_{\Delta p_{ij}}(R_i') &= \operatorname{Exp}(-\phi_i) R_i^T \left(p_j - p_i - v_i \Delta t_{ij} - \tfrac{1}{2} g \Delta t_{ij}^2\right) - \Delta \tilde{p}_{ij} \\ &\simeq r_{\Delta p_{ij}} + \left[R_i^T \left(p_j - p_i - v_i \Delta t_{ij} - \tfrac{1}{2} g \Delta t_{ij}^2\right)\right]^\wedge \phi_i. \end{aligned} \tag{136}$$

Therefore,

$$\boxed{\frac{\partial r_{\Delta p_{ij}}}{\partial \phi_i} = \left[R_i^T \left(p_j - p_i - v_i \Delta t_{ij} - \tfrac{1}{2} g \Delta t_{ij}^2\right)\right]^\wedge.} \tag{137}$$

The bias derivatives are

$$\frac{\partial r_{\Delta p_{ij}}}{\partial b_{g,i}} = -\frac{\partial \Delta \tilde{p}_{ij}}{\partial b_{g,i}}, \qquad \frac{\partial r_{\Delta p_{ij}}}{\partial b_{a,i}} = -\frac{\partial \Delta \tilde{p}_{ij}}{\partial b_{a,i}}. \tag{138}$$

17 Implementation Overview

This section briefly connects the mathematical pre-integration model with the graph-optimization implementation. It introduces no new measurement model or Jacobian derivation.

1. **Integrate IMU measurements.** For each IMU interval, accumulate $\Delta \tilde{R}_{ij}$, $\Delta \tilde{v}_{ij}$, and $\Delta \tilde{p}_{ij}$ using the fixed bias linearization point.

2. **Propagate uncertainty and bias sensitivities.** Update the pre-integration covariance Σ_{ij} and the five bias Jacobians alongside the accumulated measurements.

3. **Construct the IMU factor.** Between keyframes i and j, evaluate the nine-dimensional residual $r_{\text{pre}} = (r_{\Delta R_{ij}}^T, r_{\Delta v_{ij}}^T, r_{\Delta p_{ij}}^T)^T$ and use Σ_{ij}^{-1} as its information matrix, when invertible. Bias random-walk factors separately constrain the endpoint biases.

4. **Optimize the navigation states.** Combine the IMU factor with any available additional observation factors and optimize the keyframe states using residuals and their Jacobians. Changes in the bias estimate are handled with the first-order bias correction model (or re-integration when necessary).

In short, IMU pre-integration produces a *relative-motion measurement and its uncertainty*; the graph optimizer estimates the *navigation states*. This distinction parallels the separation between propagation and measurement update in ESKF, although the optimization mechanism is different.

18 Conceptual Summary: From IMU Measurements to Graph Optimization

The main goal of IMU pre-integration is to compress many high-rate IMU samples between two keyframes into one relative-motion constraint for graph optimization. The pre-integration computation, measurement model, and residual are not three separate estimation algorithms: they are three connected stages of the same construction.

18.1 Three distinct objects

Pre-integration computation

Accumulates the IMU readings into $\Delta \tilde{R}_{ij}$, $\Delta \tilde{v}_{ij}$, and $\Delta \tilde{p}_{ij}$. These are measurements, not optimized navigation states.

Measurement model
: Relates those accumulated measurements to the navigation states at keyframes i and j, including the effects of gravity and noise.

Residual
: Measures the disagreement between the relative motion predicted by the current navigation states and the pre-integrated measurements.

18.2 Step 1: accumulate relative-motion measurements

For each IMU sample k, the pre-integration recursively updates $\Delta\tilde{R}_{i,k+1}$, $\Delta\tilde{v}_{i,k+1}$, and $\Delta\tilde{p}_{i,k+1}$. For example,

$$\Delta\tilde{R}_{ij} = \prod_{k=i}^{j-1} \mathrm{Exp}\big((\tilde{\omega}_k - b_{g,i})\Delta t_k\big). \tag{139}$$

The corresponding velocity increment is

$$\Delta\tilde{v}_{ij} = \sum_{k=i}^{j-1} \Delta\tilde{R}_{ik}(\tilde{a}_k - b_{a,i})\Delta t_k. \tag{140}$$

The position increment is accumulated recursively using the preceding velocity increment and the measured specific force. Crucially, these calculations do not require the current world-frame estimates of R_i, p_i, or v_i. Thus changing those state estimates during optimization does not require re-integrating the raw IMU samples.

18.3 Step 2: relate measurements to keyframe states

The optimization variables at a keyframe are

$$x_k = (R_k, p_k, v_k, b_{a,k}, b_{g,k}). \tag{141}$$

The pre-integration measurement model provides the bridge between the accumulated IMU quantities and these unknown states:

$$\Delta\tilde{R}_{ij} \simeq R_i^T R_j \,\mathrm{Exp}(\delta\phi_{ij}), \tag{142a}$$

$$\Delta\tilde{v}_{ij} = R_i^T(v_j - v_i - g\Delta t_{ij}) + \delta v_{ij}, \tag{142b}$$

$$\Delta\tilde{p}_{ij} = R_i^T\big(p_j - p_i - v_i\Delta t_{ij} - \tfrac{1}{2}g\Delta t_{ij}^2\big) + \delta p_{ij}. \tag{142c}$$

Here $\delta\phi_{ij}$, δv_{ij}, and δp_{ij} are accumulated *measurement-noise perturbations*, not optimization increments applied to the keyframe states.

18.4 Step 3: construct a residual

The optimizer evaluates how well its current state estimates satisfy the IMU measurement model. Using the residual convention in this chapter,

$$r_{\Delta R_{ij}} = \mathrm{Log}\Big(\Delta\tilde{R}_{ij}^T R_i^T R_j\Big), \tag{143a}$$

$$r_{\Delta v_{ij}} = R_i^T(v_j - v_i - g\Delta t_{ij}) - \Delta\tilde{v}_{ij}, \tag{143b}$$

$$r_{\Delta p_{ij}} = R_i^T\big(p_j - p_i - v_i\Delta t_{ij} - \tfrac{1}{2}g\Delta t_{ij}^2\big) - \Delta\tilde{p}_{ij}. \tag{143c}$$

Stacking these three three-dimensional residuals gives

$$r_{\mathrm{pre}} = \begin{bmatrix} r_{\Delta R_{ij}} \\ r_{\Delta v_{ij}} \\ r_{\Delta p_{ij}} \end{bmatrix} \in \mathbb{R}^9. \tag{144}$$

For example, if the IMU-derived velocity increment is $\Delta\tilde{v}_{ij} = (2,0,0)^T$ m/s but the current states predict $(1,0,0)^T$ m/s (with $R_i = I$ and gravity removed), then $r_{\Delta v_{ij}} = (-1,0,0)^T$ m/s. The optimizer adjusts the states to reduce this discrepancy while satisfying other measurement factors.

18.5 Why covariance and bias Jacobians are needed

Pre-integration covariance. IMU measurement noise accumulates through the same integration process. Its first-order propagation is

$$\eta_{i,k+1} = A_k\eta_{ik} + B_k\eta_{d,k}, \qquad \Sigma_{i,k+1} = A_k\Sigma_{ik}A_k^T + B_kQ_{d,k}B_k^T. \tag{145}$$

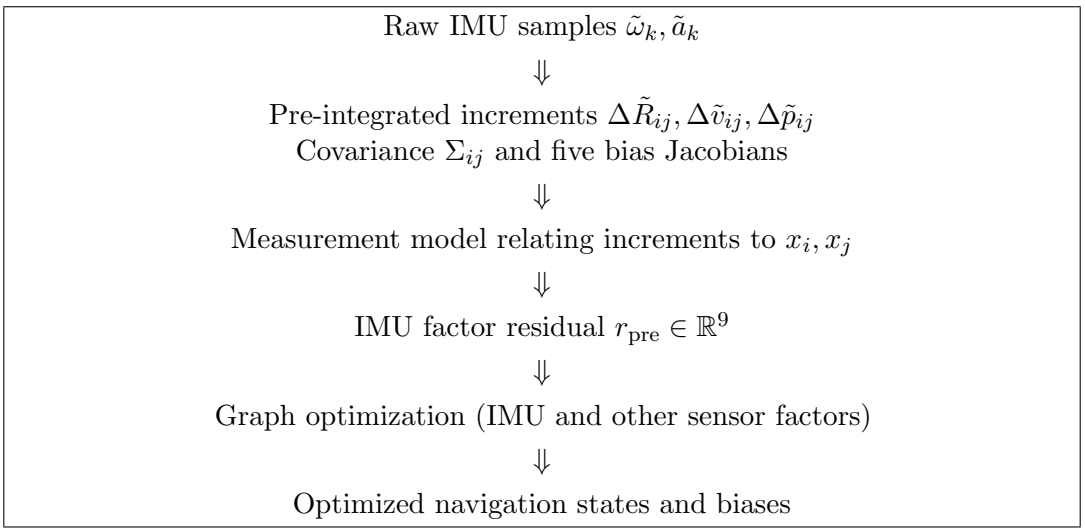
The resulting $\Sigma_{ij} \in \mathbb{R}^{9\times9}$ describes the uncertainty of the relative-motion measurement, *not* the covariance of the optimized keyframe states. It supplies the information matrix $\Omega_{ij} = \Sigma_{ij}^{-1}$ (when invertible) for the weighted residual cost

$$E_{\text{IMU}} = r_{\text{pre}}^T\Omega_{ij}r_{\text{pre}}. \tag{146}$$

Bias Jacobians. Pre-integration uses fixed gyroscope and accelerometer bias linearization values $b_{g,i}$ and $b_{a,i}$. Because the graph optimizer also estimates the biases, changes in these values alter the accumulated measurements. The five bias Jacobians provide a first-order correction without re-integrating every IMU sample at each iteration; sufficiently large changes may still require re-integration.

Residual Jacobians. These are different from both the noise propagation matrices and the bias-correction Jacobians: they describe how r_{pre} changes when the *optimization states* are perturbed, and are used to construct the graph optimizer's linearized least-squares system.

18.6 The complete picture and comparison with ESKF



In Chapter 3, ESKF uses IMU readings to propagate a navigation-state estimate and then incorporates observations through a Kalman update. Here, IMU pre-integration instead creates a relative-motion *measurement factor*; graph optimization adjusts the keyframe states jointly to reduce the weighted residuals from IMU and other sensors. The underlying IMU kinematics are closely related, but the quantities propagated and the way measurements are fused differ.

Four takeaways. Pre-integration computes relative-motion measurements; the measurement model relates them to the unknown navigation states; the residual quantifies their disagreement; and graph optimization adjusts the states to minimize the weighted disagreements.

A Complete IMU-Factor Jacobian Matrix

Use the residual ordering $(r_{\Delta R}, r_{\Delta v}, r_{\Delta p})$ and optimization perturbation ordering $(\phi_i, \delta p_i, \delta v_i, \epsilon_g, \epsilon_a, \phi_j, \delta p_j, \delta v_j)$. Each variable block has three components, so the assembled Jacobian has dimensions 9×24. For the rotation blocks, retain the shorthand

$$W := J_r^{-1}(r'), \qquad G := -W \operatorname{Exp}(r')^T J_{r,b} \frac{\partial \Delta \tilde{R}_{ij}}{\partial b_{g,i}}. \tag{147}$$

The rotation blocks below use the bias-corrected residual r'; for the other two residuals the measurement bias Jacobians are those derived in the previous section. We display the matrix in three 3×24 block rows to avoid a page-width overflow.

$$\begin{aligned}
J_{ij} &= \frac{\partial r_{ij}}{\partial \delta X} = \begin{bmatrix} J_R \\ J_v \\ J_p \end{bmatrix}, \\
J_R &= \begin{bmatrix} -W R_j^T R_i & 0 & 0 & G & 0 & W & 0 & 0 \end{bmatrix}, \\
J_v &= \begin{bmatrix} \left[R_i^T (v_j - v_i - g\Delta t_{ij})\right]^\wedge & 0 & -R_i^T & -\frac{\partial \Delta \tilde{v}_{ij}}{\partial b_{g,i}} & -\frac{\partial \Delta \tilde{v}_{ij}}{\partial b_{a,i}} & 0 & 0 & R_i^T \end{bmatrix}, \\
J_p &= \begin{bmatrix} \left[R_i^T (p_j - p_i - v_i \Delta t_{ij} - \frac{1}{2} g \Delta t_{ij}^2)\right]^\wedge & -R_i^T & -R_i^T \Delta t_{ij} & -\frac{\partial \Delta \tilde{p}_{ij}}{\partial b_{g,i}} & -\frac{\partial \Delta \tilde{p}_{ij}}{\partial b_{a,i}} & 0 & R_i^T & 0 \end{bmatrix}.
\end{aligned} \tag{148}$$

Here every displayed block is 3×3. The position and velocity rows do not directly depend on R_j in this residual convention. The Jacobian above is evaluated at the current bias-corrected pre-integration. The velocity and position bias derivatives denote the stored first-order sensitivities.

A.1 Scope and next steps

The present notes stop at the position residual Jacobians. The assembled matrix is a convenient grouping of the individual Jacobian blocks, rather than a separate equation in the textbook. No additional sensor factors are included here.

A.2 Full two-keyframe state ordering

The displayed 9×24 matrix uses only the eight three-dimensional state blocks directly connected to the pre-integration factor. For the complete two-keyframe ordering

$$\delta X_{ij}^{\text{full}} = (\phi_i, \delta p_i, \delta v_i, \delta b_{a,i}, \delta b_{g,i}, \phi_j, \delta p_j, \delta v_j, \delta b_{a,j}, \delta b_{g,j}), \tag{149}$$

the full Jacobian has size 9×30. It is obtained by reordering the existing blocks to match this ordering and appending two zero 9×3 blocks for the endpoint biases. Those endpoint biases are constrained by the separate bias random-walk factors.